

THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

A3 BOOKLET

Haidian District, Beijing • both sides of the Jing-Zhang Rail Heritage Park • three key areas: Zhongzhiyuan • Beijing AI Origin Community • Dazhongsi AI Industry Cluster

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.

Depart, carry the height station by station, return — that gap is f.

Within tolerance, every station on the line is accepted;

over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:

what a city's trust in its AI rests on: not performance, but return.



Ground within a ten-minute walk of a benchmark

Dashed: 183 buildings today. Filled: the ground that only comes within ten minutes once the 11 stitching links and three spurs are built — 206 in all, a gain of 23.

Catchment 164.1 ha to 178.4 ha; 720 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

11,412,825 m²

Overall design area

9,443 m

Spine length

8

Tiered benchmarks

13.8 km/km²

Network density, measured

17

Built-up oversized blocks

19.7%

Green corridor, share of design area

What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

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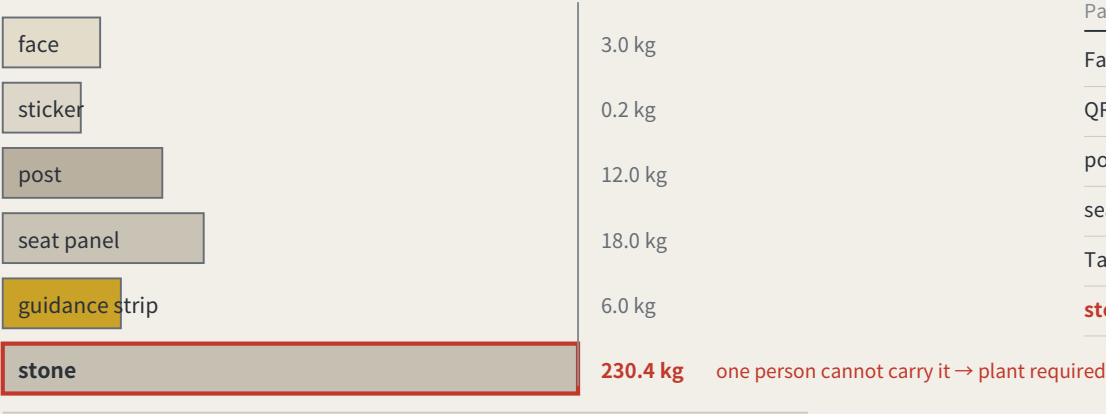
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Maintenance: putting the wear on the parts that can be swapped

FIG.18

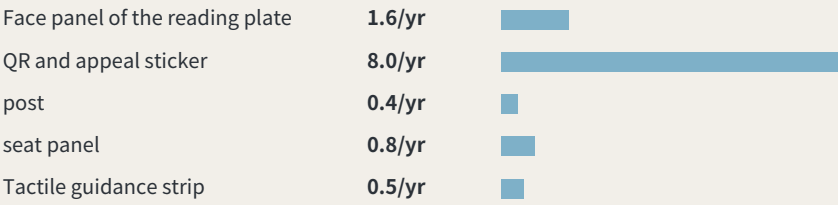
a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved



IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
stone	impact, excavation, settlement	never replaced	230 kg	plant required	yes

III. After it fails: found → held → replaced → lost

Found	any re-survey that finds the face stale or missing records it in that cycle the plate’ s own rule — a stale value counts as not posted — is the detector
Held	the point is marked returned in datum red, shown at every third-order point in the area bad news belongs where it happened (errata E50)
Replaced	a standard part is swapped without touching the stone or its mark centre one person, one visit; heaviest part 18 kg
Stone lost	the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused reusing an id would quietly join two different points into one history

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

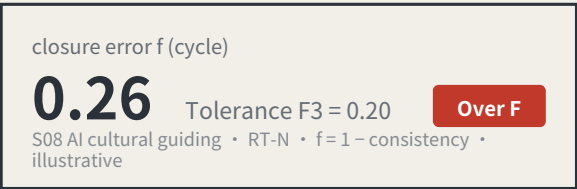
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

L3 return-to-datum point

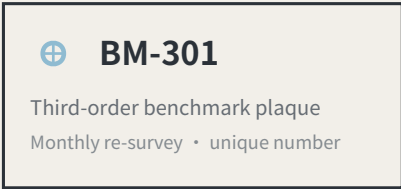
BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

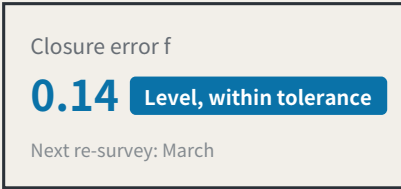
Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language



1 Stone and plaque

The number is position, order and cycle



2 Reading board

Visible on site; no internet needed



3 Dwellable seating

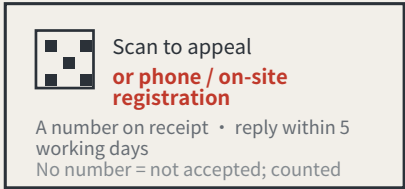
With armrests: older people rely on them to stand



Clear width and gradient enter the re-survey items

4 Step-free guidance

Read by the users themselves



5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	2nd-order point	Quarterly re-survey
BM-3xx	3rd-order point	Monthly re-survey
RT-N / RT-S	Connecting route	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	Community day	third order BM-301/BM-302/BM-303	P4 • P5 • P7
Quarterly	Scenario open day	second order BM-21/BM-22	P2 • P3
Semi-annual	Route re-survey	first order, routes RT-N / RT-S	P1 • P2
Annual	Zeroing ceremony	L3 return-to-datum point	all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

Method, two measurements, and the boundaries of application

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Closure error in the site's own data

412.5 m	Provisional design area to surveyed park
100%	Share lying inside the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level contains it wholly; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
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Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

A city that publishes its own error.



FIG.01

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

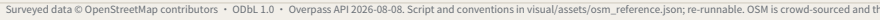
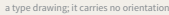


FIG.02



MEASURING THE OPEN CALL ITSELF • as of 2026-08-21 • 939 merged • 939/939 fetched • PR numbers past 1000

The gallery index lists 872 at the same moment — 67 behind

164 of 939 still hold the scaffold placeholder or are empty

The GPT/Codex family alone is written 93 ways across 413 proposals

The fix is light: split `model` into an enumerated family plus free-text detail, and add one enumeration check.

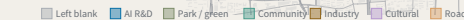
Registers answer “what did the system declare” and assessments answer “what do experts think”; closure error answers “how much does the conclusion differ across nodes and across time for the same public question”

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-21 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

FIG.03

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



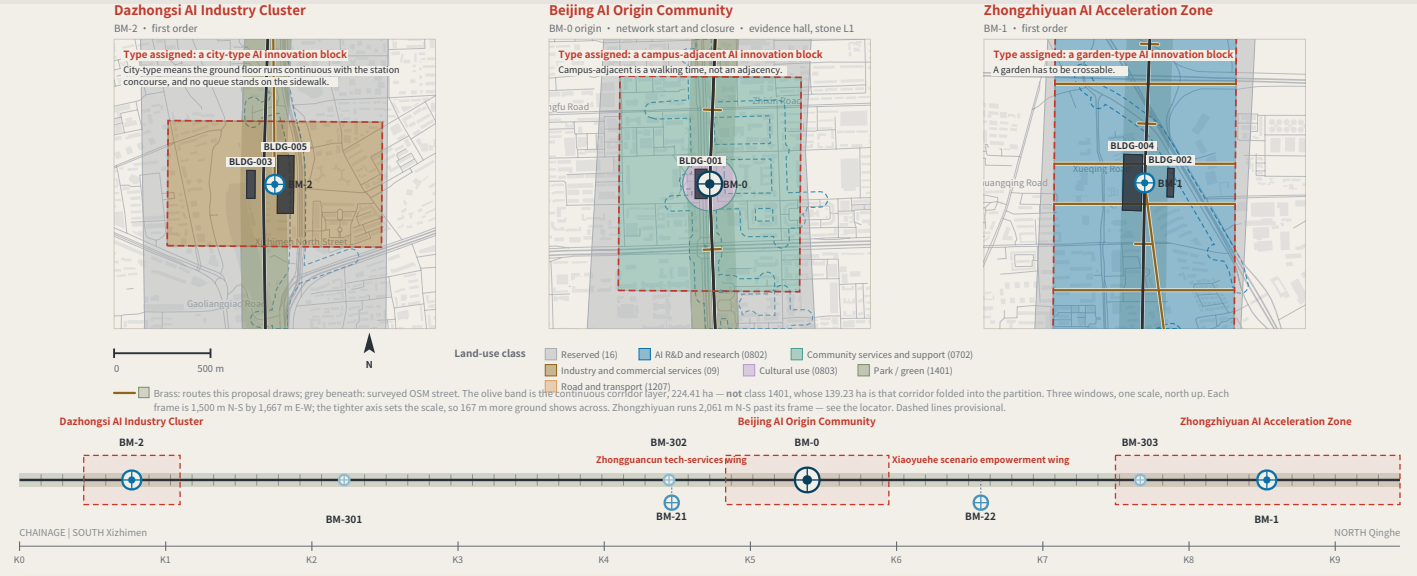
Any area, ratio or count that cannot be recomputed from a structured layer is not written as a conclusion.

THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

FIG.04

Each area and wing carries a survey role — origin, tolerance, density; the blue dashes are the ten-minute walk catchment, by network distance, not radius.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Benchmark orders



Closure verdict

CLOSURE RULE

A scenario departs BM-0, is read at each station along a connecting route by a different party, and terminates at a first-order point; the computation closes at BM-0.

● $f \leq F$ level for this cycle: it may operate and enter the next cycle

● $f > F$ the whole route returns for re-survey and drops to the non-AI equivalent

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

Layers: geometry/key_areas.geojson, roads.geojson, public_space.geojson, green_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). In all three windows the base street network and building footprints are measured OpenStreetMap data (Odb: 1.0, the same cache as the density readings), drawn for location only and not part of this proposal's design. The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-21 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Three areas, two wings and survey roles

Dazhongsi AI Industry Cluster

High-frequency reading point | AI-native consumption and business

Beijing AI Origin Community

Network origin and closure computation | public evidence hall and stone L1

Zhongzhiyuan AI Acceleration Zone

Tolerance datum | where F is set

Zhongguancun tech-services wing

Support system | capital and factors; takes no reading of its own

Xiaoyuehe scenario empowerment wing

Where the stations are densest | real users

A wing carries a second-order point, not an area design; its requirement is written as dimensions, on FIG.26.

Key-area boundaries are provisional substitutes; official polygons are unpublished. Dashed; not a redline or an area basis.

BM-2 • first order

BM-0 origin benchmark

BM-1 • first order

BM-21 second-order benchmark

BM-22 second-order benchmark

RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

METRICS RECOMPUTATION

RECOMPUTED METRICS • EPSG:4548

Class 1, measured: recomputable directly from this package's geometry

Overall design area	11,412,825 m ²	S06 low-speed device pilot	237,479 m ²
Spine length	9,443 m	S10 public-safety review	15,369 m ²
Benchmark count	8	Green ratio (land-use partition basis)	0.1220
Green corridor ratio	0.1966	Phase increments, summed	2,920,311 m ²
Public measurement-point ratio	0.0642	Phased share of design scope	0.2559
Indicative footprint area	82,276 m ²	Existing street length, measured	157,510 m
Green corridor area (footprints deducted)	2,244,082 m ²	Existing network density, m/m ²	0.0138
Public measurement-point area	733,011 m ²	Existing median block, measured	4,300 m ²
Key areas	3	Existing building coverage, lower bound	0.1798
Key areas, total	3,692,893 m ²	Six spine benchmarks walked, mapped net	11,472 m
S11 industry test ground	53,008 m ²		

A further 22 count metrics (unit: count) ship in metrics.json:

cards, personas, landmarks, cases, projects, phases, land-use classes, errata — all computed by the build, none in this table.

Class 2: awaiting official controls or a field survey, held at unknown

Floor area ratio • Building height • Building density • Setback (incl. road redline) • Existing residential floor area

Filling estimates into a gap while official data is absent is fabricated certainty.

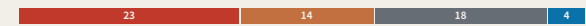
Class 3: requires continuous re-survey calibration; no baseline yet

Scenario closure error f • Tolerance pass rate • Non-AI equivalent path coverage • Resident-led remeasurements

Baselines follow one near-term cycle. This proposal says no data exists rather than passing intent off as measurement.

Evidence ceiling: computed exactly $\neq$ measured reliably

Each metric's ceiling is the weakest of the files it reads, computed by the build rather than judged by the author



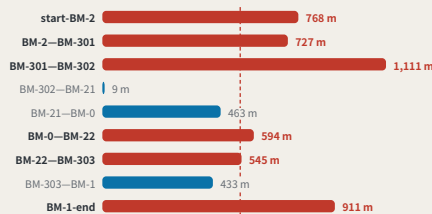
60 metrics, 59 of them gradeable; 23 are drawn on an inferred boundary and 0 reach the regulatory baseline.

Every class-1 value is recomputed from the shipped layers by the accompanying script;

A number nobody can recompute is not evidence — and that standard applies to this proposal first.

Walk to the nearest benchmark: 9 segments, 6 of them failing

WALK TO THE NEAREST BENCHMARK • worst per segment; red line 540 m (PS at 0.6 m/s, 15 min)



Worst 1,111 m; for PS that is 30.9 minutes - this proposal's own rule, and the layout as drawn does not meet it. Closing it needs 9 more third-order points, whose annual operating cost is already in this package's cost table rather than left to a next version.

Sources: this package's metrics.json and visual/assets/field_map.json, both shipped with it; the authoritative source is the GitHub git tree. The independent verifier visual/assets/verify.js checks every class-1 metric. Generation scripts are in the accompanying issue. The corpus grows daily; re-run before citing.

THE LEVELING LINE • Haidian, Beijing • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-21 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Closure evidence and measured corpus

CLOSURE EVIDENCE AND MEASURED CORPUS • n=939 (git tree) • 2026-08-21

Closure evidence: the mechanism has been run

node visual/assets/run_s08_tabletop.js • 10 cases • 2 accepted • 8 refused • re-runnable by the reviewer • rule and reason quoted verbatim from the shipped file

accept	$f \Leftarrow F$ is level for this cycle and the scenario continues
accept	$f = F$ returns the whole route, not the failing station
refuse	rule 5: amending only the worst station is not an available action
refuse	record: stations: needs at least 3 items, has 2; review parties: 2 distinct across the circuit; a reading taken by one kind of party cannot detect what that kind of party cannot see
refuse	a record does not get to state its own f
refuse	verdict: f declares 0.13 but the stations recompute to 0.360000; verdict: passed says true but 0.360 against F 0.2 derives false; verdict: action is "continue" but against F requires "return-whole-route"
refuse	a circuit read by one kind of party cannot detect what that party cannot see
refuse	review parties: 1 distinct across the circuit; a reading taken by one kind of party cannot detect what that kind of party cannot see without an equivalent non-AI path the stop rule would be circumvented
refuse	record: verdict: non_ai_path_available: must be true, got false; verdict: non_ai_path_available is false: a scenario with no non-AI equivalent cannot be returned, because returning it would interrupt public service — so the stop rule would be circumvented
refuse	omitting the non-AI path field is not the same as satisfying it
refuse	record: verdict: missing required property "non_ai_path_available"; verdict: non_ai_path_available is absent: the equivalent non-AI path is not waivable, so omitting the field cannot be treated as satisfying it. State it explicitly as true.
refuse	a connecting route must terminate at a first-order benchmark, or nothing checks the carried value
refuse	station(s): 4: a connecting route must terminate at a first-order benchmark, because the carried value has nothing to be checked against otherwise. This one ends at BM-302 (third).
refuse	rule 7: two consecutive passes are required, because once may be luck
refuse	resumption needs two consecutive passing cycles, record shows 1; once may be luck
refuse	rule 7: an F1 incident resumes only on unanimous confirmation, no majority rule
refuse	resumption after an F1 incident requires unanimous confirmation by all four review categories

Nothing here is a field measurement. Not one reading was taken: these are worked values chosen to exercise the refusal branches.

Measured corpus: few take this road

939 packages, read from the git tree (listed in the index). Pulse or lifetime is at 195 and proof or verification at 210, both saturated; measurement or levelling is at 13. Most detection is a keyword regex and therefore a lower bound, and the convergence on three-areas-two-wings is induced by the taskbook rather than agreed.

Leveling is not another railway-operations metaphor. It is a method for how trust is established.

What is left of the belt with the AI switched off

Without a non-AI equivalent the stop rule is circumvented. All 12 cards name their equivalent; the 4 F1 ones keep theirs permanently (full text in scenario_cards.json).

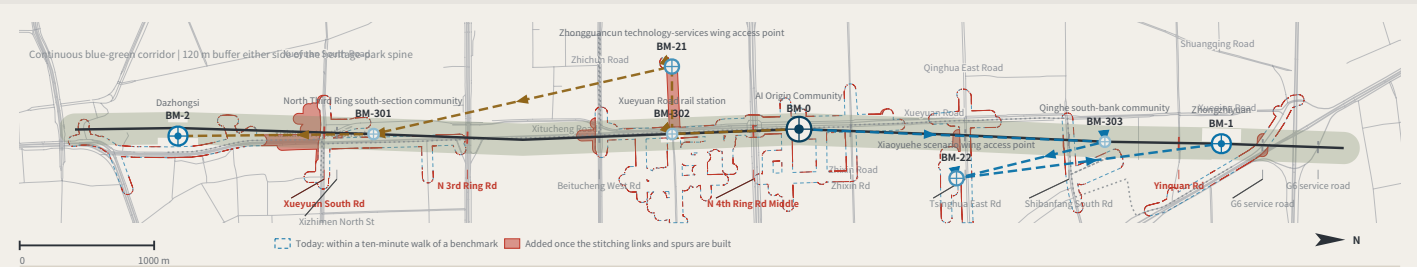
S01 F3	Paper sign-in and on-site stewards	S02 F2	Residents report breaks on paper at the nearest third-order benchmark...	S03 F2	The staffed counter runs independently of the agent
S04 F1	A staffed help desk and a printed directory of institutions, departments...	S05 F2	Authorisation credentials and change records are documents	S06 F1	Devices stop and return to their bays
S07 F3	The evidence hall's physical display and paper archive stay open	S08 F3	Physical interpretation plates along the route and scheduled human-led tours	S09 F2	Prices and licences go on a physical notice board
S10 F1	A review meeting and a paper record	S11 F1	The test ground stops serving the public and its boundary is held by...	S12 F2	Checking reverts to user reports on paper or by phone plus quarterly...

SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 183 to 206.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



East-west stitching points (11)

EAST-WEST STITCHING POINTS

The class rests on one measurable fact: whether OSM already maps a crossing within 60 m of the intersection. If it does, Class A. If it does not, the need is registered and left ungraded — separating Class B (channelisation) from Class C (new structure) is a traffic study's conclusion, not this proposal's.

Yinquan Rd	Class A	10 m	Xueyuan South Rd	Class A	25 m	N 4th Ring Rd Middle	Class A	30 m	N 3rd Ring Rd	Class A	33 m
Beitucheng West Rd	need registered only	65 m	Xizhimen North St	need registered only	68 m	Tsinghua East Rd	need registered only	86 m	Shibanfang South Rd	need registered only	146 m
G6 service road	need registered only	166 m	G6 service road	need registered only	175 m	Zhixin Rd	need registered only	201 m			

Distances are to the nearest mapped crossing. OSM is crowd-sourced: "need registered only" means OSM shows none, not that none exists on the ground.

The two connecting routes

CONNECTING ROUTES

North connecting route RT-N

BM-0 → BM-303 → BM-22 → BM-1 (terminates at a first-order point; the computation closes at BM-0)

South connecting route RT-S

BM-0 → BM-302 → BM-21 → BM-301 → BM-2 (terminates at a first-order point; the computation closes at BM-0)

Walking the six spine benchmarks on the mapped network, measured

11,472 m against 9,443 m, 2,029 m further today, x1.21 — that difference is what the belt is for

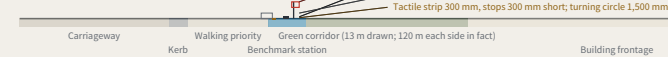
Two of the eight are outside this chain. BM-21 has no walkable path to any other benchmark on this graph at all — not a detour, unreachable. BM-22 is reached by a spur and is registered separately, so that one leg is not counted twice.

Each station is read independently by a different party — professional body, operator, resident representative, international visitor, all four required. Homogeneous review parties would bias the closure error systematically small and rob it of meaning, a risk this mechanism declares.

Spine section, at the scale of a person

SPINE SECTION AT THE SCALE OF A PERSON

On A0, about 1:100; from KIT-01 to 04, no vertical exaggeration



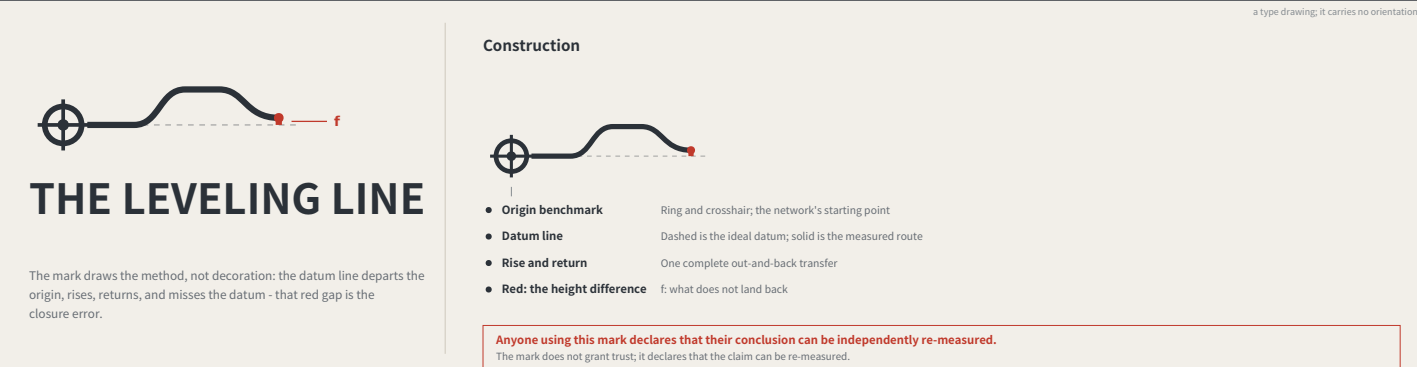
Layers: geometry/roads.geojson, green_space.geojson, public_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). Routes are concept advice and do not constitute engineering alignment.

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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

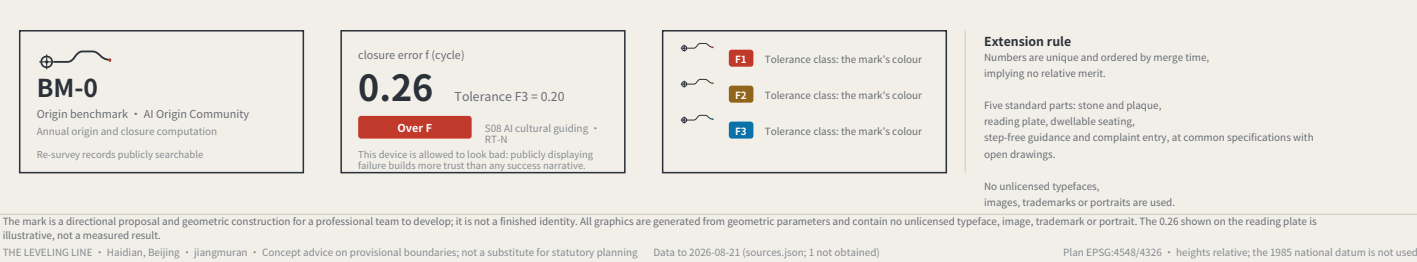
FIG.07



VARIANTS



APPLICATIONS



a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun tech-services wing	→	BM-21	→	Deliberately blank – see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Numbering	Case	Trust mechanism	Re-measurable	Transferable point
C1	Algorithm registers	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	Risk-tiered AI legislation	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	Standardised testing frameworks	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	Algorithmic impact assessment practice	Mandatory ex-ante and ex-post review for high-risk uses	Red-hip	Supports the strictest tolerance for health scenarios
C5	Civic data-stewardship practice	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	Open-source reproducibility norms	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied; no marks used; no non-public data; and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land-use, geojson and public_space.geojson.

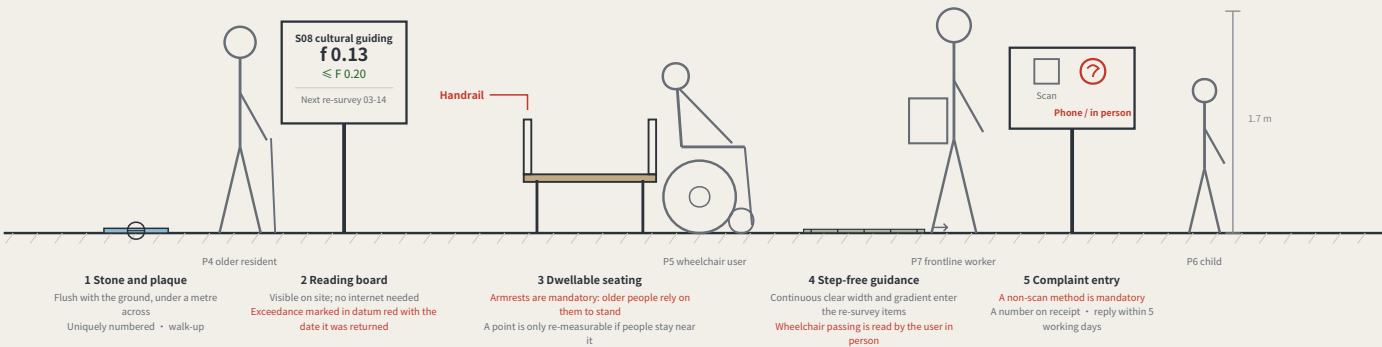
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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

What a benchmark looks like at eye level

A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

The same kerb, after snow clearance

In winter the order becomes one real trade-off

Piled on the device band

Devices stop; S06 falls back to staffed transfer

Piled on the pedestrian band

Pedestrians pushed into the carriageway - the worst of the three

Piled on the kerbside parking band

The one item in the order that may be lost first

This proposal takes this one

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths



satisfied first →

→ allocated last

The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. The plate's f 0.13 is an illustration, not a reading: none is measured yet. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.

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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

An executable form of coordination: how two levelling networks join



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named.

No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

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a type drawing; it carries no orientation

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY

2. Dimensions: design intent / as-measured

DIMENSIONS: DESIGN INTENT vs AS-MEASURED

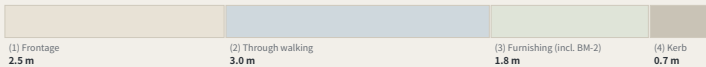
Item	Design intent	As measured
Pedestrian holding area, per quadrant width $\approx$ 3.0 m x depth $\approx$ 4.0 m	$\geq 12 \text{ m}^2$	
Crossing effective clear width after deducting poles, covers and the furnishing strip	$\geq 4.0 \text{ m}$	
Kerb ramp gradient Barrier-Free Environment Construction Law and current codes	$\leq 1:12$	
device queue reservoir set behind the building line, with its own kerb cut	$\geq 18 \text{ m}^2$	
Reservoir to holding area, clear distance must not cross any desire line	$\geq 2.0 \text{ m}$	
BM-2 stone to pedestrian edge the stone is flush with the ground and is not a trip hazard	$\geq 0.6 \text{ m}$	

The right-hand column is deliberately empty.

Capacity is computed on site from measured clear width. A number filled in without measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION



Building line to kerb: frontage, walking, furnishing, kerb.BM-2 sits in the furnishing strip, so a reading is never taken in the walking zone.



Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut.

If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

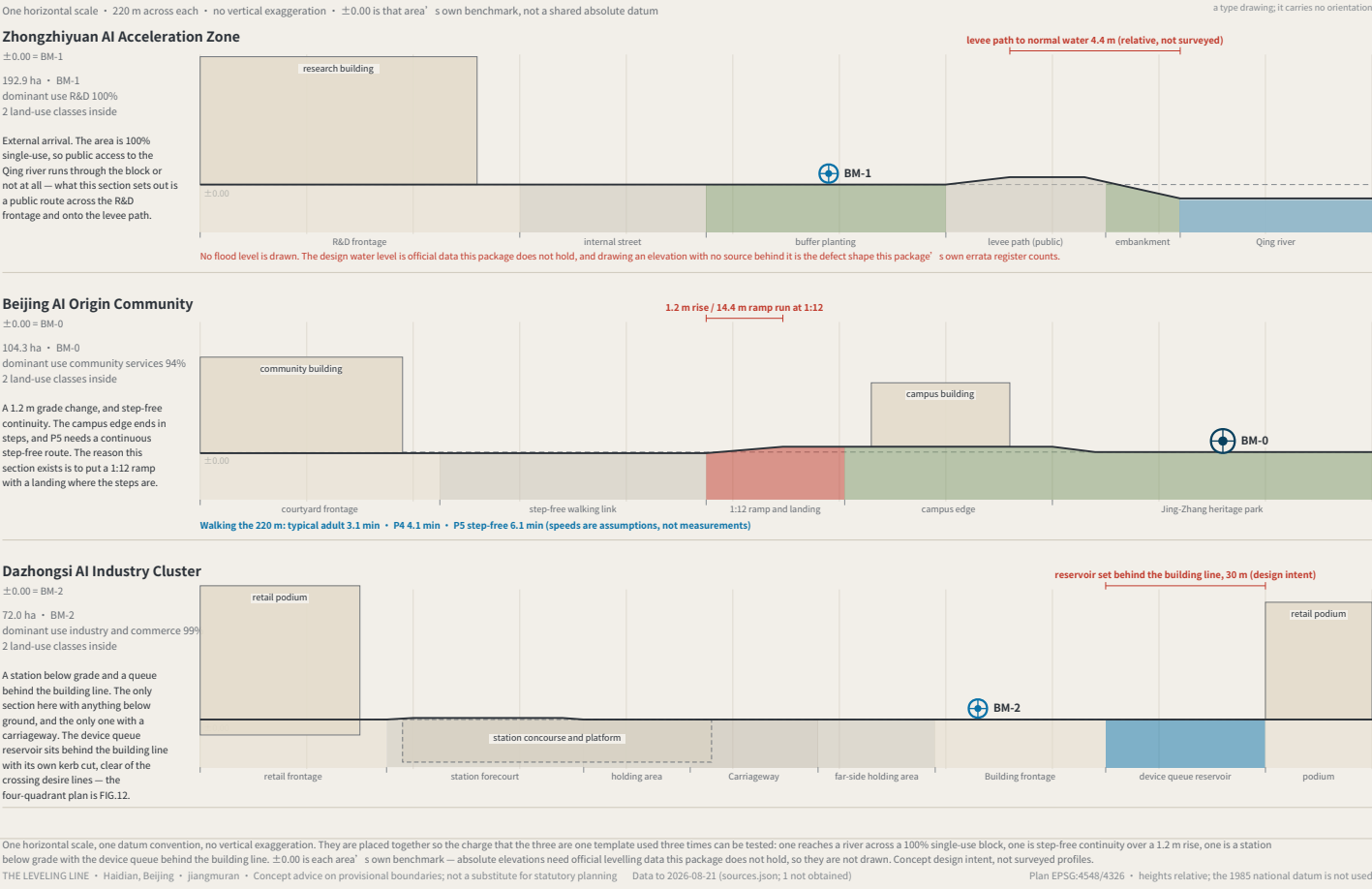
This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.

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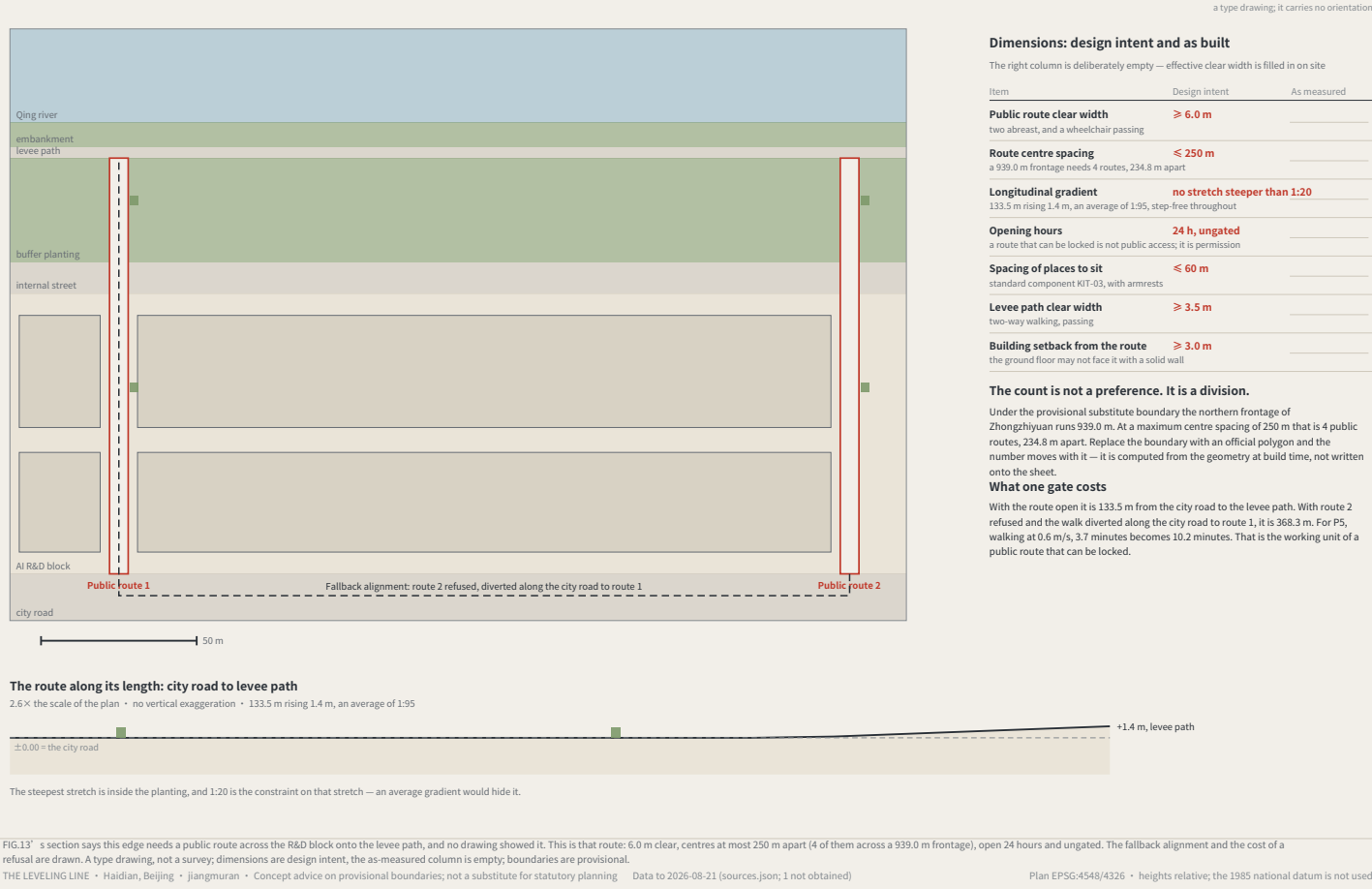
The three key areas in section, at one scale

FIG.13



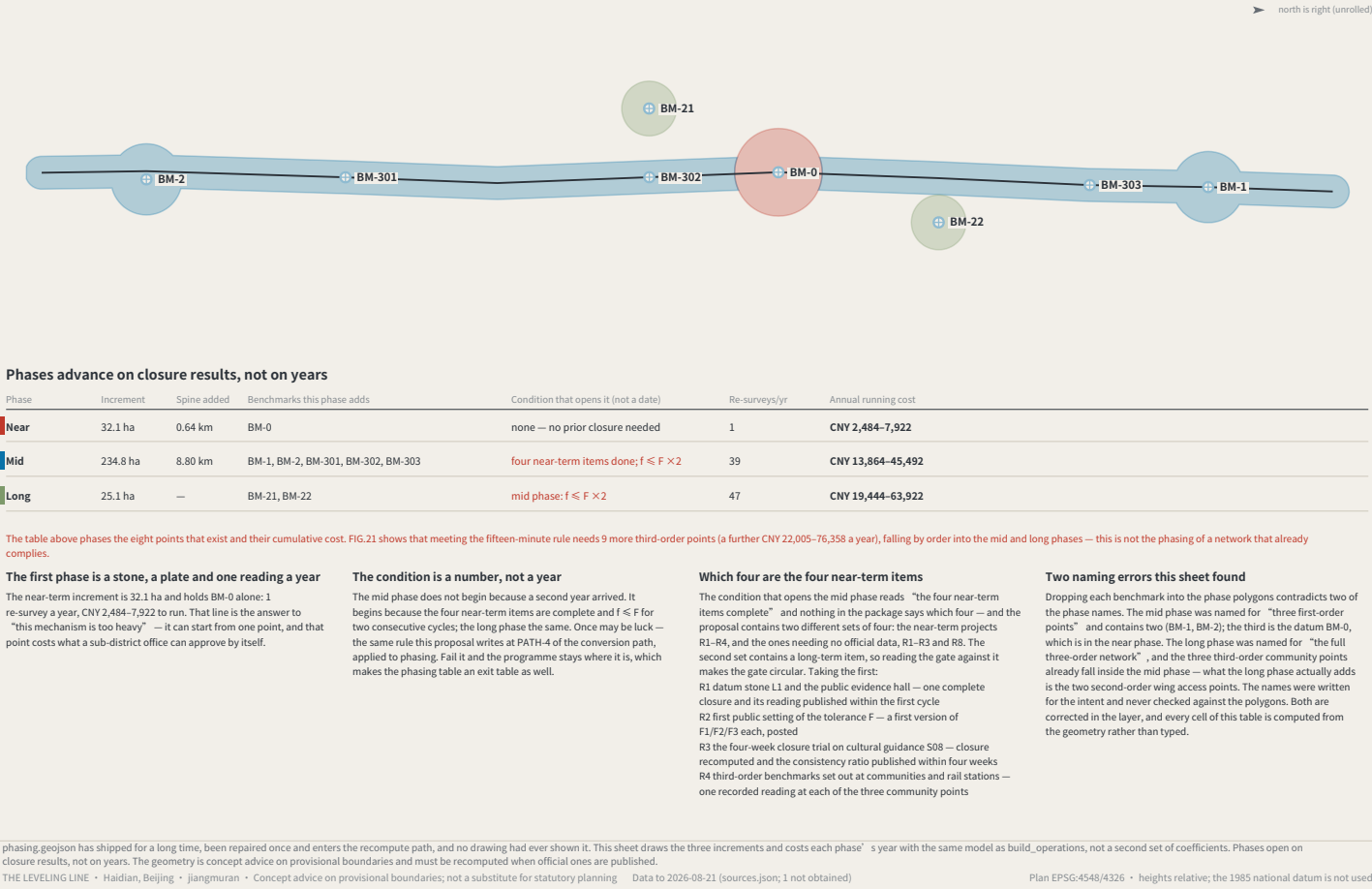
The through-block public route to the Qing river

FIG.14



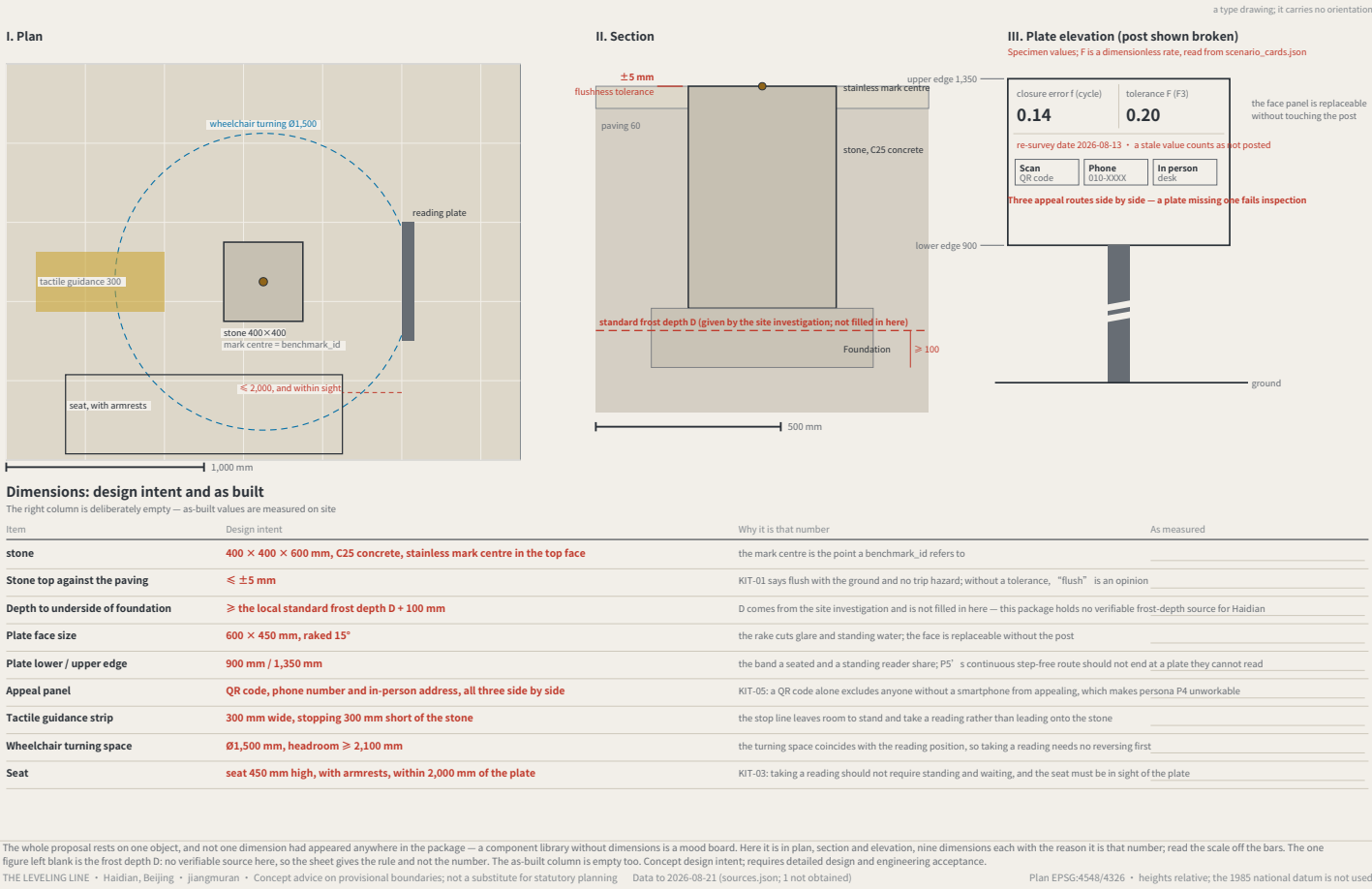
Phasing: advanced by closure results, not by dates

FIG.15



The benchmark and its reading plate, as a construction detail

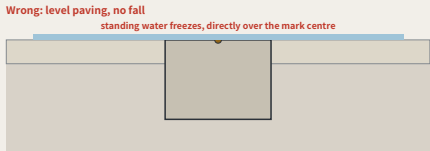
FIG.16



Winter: water, ice, and two of this package's rules in collision

FIG.22

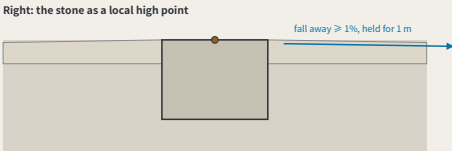
I. Two ways of doing it, at one scale



Level paving with no fall: compliant in September, the only ice in the sidewalk in January

1,000 mm

The fall is drawn true: 1% over 1 m is 10 mm, which is barely visible at this scale — and a slope nobody can see on a drawing is a slope nobody checks on site, while every dimension on the sheet is still satisfied.



The stone is the local high point: flush with the paving beside it, and that paving falls away

II. What that requires of the construction

Item	Requirement	Why
Stone top against the paving	flush, tolerance ± 5 mm	KIT-01: no trip hazard
Fall around it	$\geq 1\%$, away from the stone, held for ≥ 1 m	makes the stone a local high point — water leaves instead of gathering
Protective ring	none	a recessed ring is a bowl; a raised one is the trip hazard KIT-01 forbids
Surfacing	permeable or tight-jointed; no joint channel running inward	a joint channel leads water to the mark centre
Winter reading	taken after clearing, with the condition recorded	a cleared reading and an uncleared one are not directly comparable

III. Three things a winter reading adds

Clear it	clear snow and ice from the stone and 300 mm around it before reading a reading taken over a film of ice is measuring the ice
Check the drainage	confirm the stone is still the high point: no ponding marks or ice within 1 m a ponding mark is evidence the drainage has already failed, and it appears before the ice
Record the condition	mark the cycle record as a winter condition, and whether the stone was cleared cleared and uncleared readings are not directly comparable — unrecorded, nobody knows whether they should be

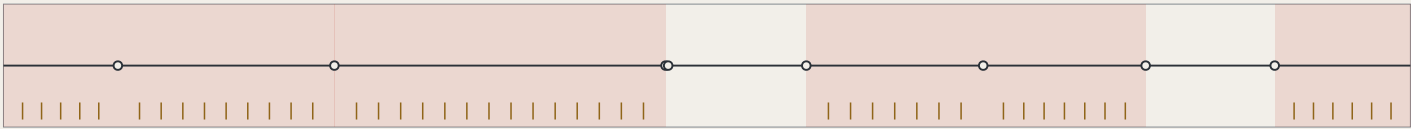
The three third-order points are read monthly, January included, and this package had never said what that costs. KIT-01 requires the stone flush with no trip hazard and FIG.16 gave that ± 5 mm — but a mark flush in level paving sits in standing water: the tolerance that makes it safe in September makes it dangerous in January. The answer is a local high point, not a ring.

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Finding the nearest benchmark: which way, and how far

FIG.23

1. Wayfinding only in the segments where this proposal is already in breach



Red marks the segments FIG.21 judged to fail the fifteen-minute rule; one wayfinding post every 150 m, 48 in all. Compliant segments get none — putting up a sign where it is already close enough is decoration.

1,000 m

2. The four things the plate must answer

← BM-302 410 m • P5 approx. 11 min	Direction ← BM-302 / BM-0 → Either end of the line may be nearer; only at the midpoint are both sides equally far
BM-0 → 520 m • P5 approx. 14 min Standard part KIT-04, no new structures	Distance 410 m / 520 m metres, because metres can be checked by the person who walks them
	Time P5 approx. 11 min / 14 min At 0.6 m/s, the same figure as in every other drawing in this package (personas.json)
	Numbering BM-302 • BM-0 A reading taken on arrival is attributed by it; no need to ask afterwards where the person was standing

3. What wayfinding cannot do

These 48 wayfinding posts will not make 1111 m any shorter. What they do is let a person know before setting out how far it is, which way, and what the point is called on arrival. This is mitigation, not a fix. The fix is the 9 third-order points in FIG.21, which give no position until the official alignment is published. Writing the wayfinding up as the solution is exactly the kind of substitution this package has kept catching in itself — using one small thing it can do to cover one large thing it cannot.

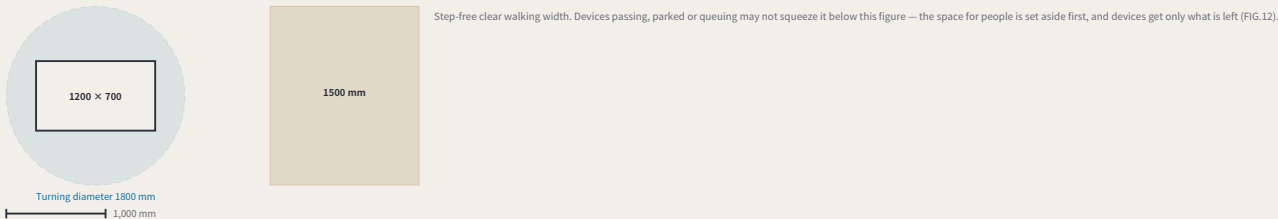
FIG.21 measures the farthest point on the whole line at 1111 m from the nearest benchmark: 9 segments, of which 6 fail this proposal's own fifteen-minute rule. Closing the gap takes 9 third-order points, but their positions await the official alignment. People still have to find the points that already exist. This drawing supplies wayfinding: placed only inside the segments already in breach, one every 150 m, 48 in total, using the existing standard part KIT-04, adding no new structures. Wayfinding will not make 1111 m any closer — this is mitigation, not repair.

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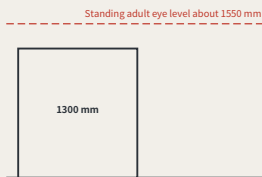
The device envelope: what exactly goes in those 18 m²

FIG.24

1. Envelope and turning, plan



2. Elevation: why the height is this number



Height sits below eye level so queued devices do not cut the sightline across the sidewalk. No external basis; set by this proposal — eye height varies from person to person, and the sheet gives the reasoning, not a clause to copy.

3. The envelope items, and the constraint source of each

Item	Value	Constraint source	Why it is that number
Length × Width	1200 × 700 mm	published by this proposal	Back-calculated from the standalone kerb cut width in FIG.12: a device that cannot pass the opening is not accepted by this proposal
Height (with load)	1300 mm	Set by this proposal	Taken below the sightline of a standing adult, so that queuing devices do not block eye contact across the sidewalk; no external basis, set by this proposal
Turning diameter	1800 mm	Set by this proposal	Required to turn in place; it sets the shape of the reservoir space and not just its area — precisely the half that "18 m ² " cannot state
Kerb weight	≤ 120 kg	Set by this proposal	Taken at the order a single person can push clear after a power failure; beyond that, mechanical assistance is required, and there is nowhere to place it at a kerb cut
Site speed	≤ 6 km/h	Set by this proposal	Speed ceiling in the shared pedestrian-vehicle zone, taken at the order of walking speed; no citable local regulation was obtained, so it is registered as a self-set value
Clear width left after passing	≥ 1500 mm	published by this proposal	Step-free clear walking width, the same value as KIT-04; equipment passage may not press it below this number
Marking	Cap height ≥ 25 mm	published by this proposal	ID number, responsible body and appeal channel, all three present; same rule as the reading plate: attribution starts with a visible ID number (FIG.16, FIG.20)

4. Converting 18 m² into a number of units — the two drawings must agree with each other

One standing bay is the envelope length times the turning circle: 2.16 m². FIG.12 publishes a 18 m² floor for the reservoir, so that is 8 at once. The number did not exist before: "≥ 18 m²" never said how many, and two devices of equal footprint with different turning radii do not need the same reservoir. The build checks the two sheets against each other — if the area does not divide into bays, or the count exceeds what the floor holds, it refuses.

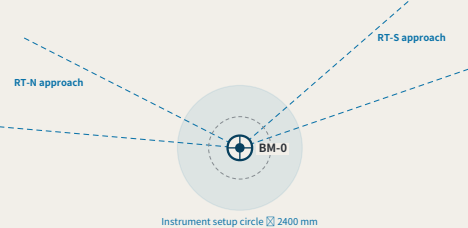
FIG.12 fixed the device reservoir at ≥ 18 m² and this package had never drawn a device. Area is not specification: equal footprints with different turning radii need different reservoirs. An envelope, not a product; 7 dimensions each marked with what bounds them, 4 bounded by nothing but this proposal — attack those first. It holds 18 m² / 8 devices.

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BM-0, the origin: the place both routes have to return to

FIG.25

1. Datum site: instrument setup circle, clear walking width, approach sightlines of the two routes



2. Closure record: readable without

A reading is published the moment it is taken, yet the evidence is public only as far as the door. So the closure record sits outside the door, facing the sidewalk, on the plate geometry FIG.16 fixes (eye height 900–1350 mm). Inside the hall it would turn anyone can walk up and read into during opening hours.

3. The six requirements the datum must meet, and the source of each

Item	Requirement	Why, and where it comes from
Instrument setup circle and the step-free clear width	Diameter 2400 mm ≥ 1500 mm clear outside the setup circle	Tripod spread 1,200 mm, operator outer circle 600 mm; closing a route means setting the instrument up on this stone, and this circle is the ground it actually takes Instrument setup may not eat the clear width — FIG.24' s rule for equipment, no slack for our own people
Route intervisibility	One approach sight cone each; no trees or parking inside	RT-N and RT-S terminate here, which needs sight of the previous turning point; without it you add one, and each adds a height difference
Closure record orientation	Faces the sidewalk, eye height 900–1350 mm, read without entering	Published as taken, but evidence is open only as far as the door — a datum with opening hours is shut when the reading lands
Display content	Network closure error $\geq 1\%$ out	Network closure error only; a single-point failure shows there (erratum E50). Moving bad news back to the datum is the recorded defect
Fall	$\geq 1\%$ out	As FIG.22: the stone must be a local high point, or in January it is the only ice here

BM-0 appears on nearly every sheet here and had never been drawn — and what it carries is a spatial property: a reading is published the moment it is taken, and evidence is only as public as the door. Hang the record in a hall with opening hours and the origin is shut at the moment the reading lands. This sheet gives the setup circle, an approach sightline cone for each connecting route, and the closure record outside the door facing the sidewalk.

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BM-2x, the second order: a scenario has to be stoppable, and that takes ground

FIG.26



3. Where second-order and third-order points differ

Item	Value	Why
stone	Same KIT-01 as the third-order point	Second order and third order differ in re-survey interval and route role, not in construction — a more important place gets no special benchmark
Re-survey cycle	Quarterly	The two wings are the source of scenario density; readings are taken quarterly, priced on the same table as the annual first order and the monthly third order
Observation front	30 m	The stretch of kerb the scenario is actually watched from. "The scenario takes its reading here", given without a length, is the same kind of sentence as "reservoir space 18 m" given without devices
Halt distance	2.59 m	From FIG.24: a site speed of ≤ 6 km/h gives reaction 1 s over 1.67 m, then deceleration at 1.5 m/s ² over 0.93 m
Clearance requirement	No fixed objects and no waiting area within ≥ 2.59 m ahead of the front	A scenario that cannot stop is not a scenario, it is an accident; this stretch of ground is the physical form of the words "it can stop"
Clear walking width	≥ 1500 mm, uninterrupted throughout	The same number as in FIG.24 and FIG.25 — observation, instrument setup, equipment passage: none of them may eat into it

The two wings carry the second-order points; they appear in the naming table, the cost table and the resource table, and have never been drawn. What separates second order from third order is not the stone — both are KIT-01 — but that what runs here is a **scenario aimed at real users**, so it must be able to stop. Computed from FIG.24 at ≤ 6 km/h, the halt distance is 2.59 m; within that stretch of ground ahead of the front there may be no fixed objects and no waiting area. The reaction time and the deceleration are both set by this proposal and are registered as A-DEVICE-002.

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Third order: the most numerous tier, and more than one reader

FIG.28



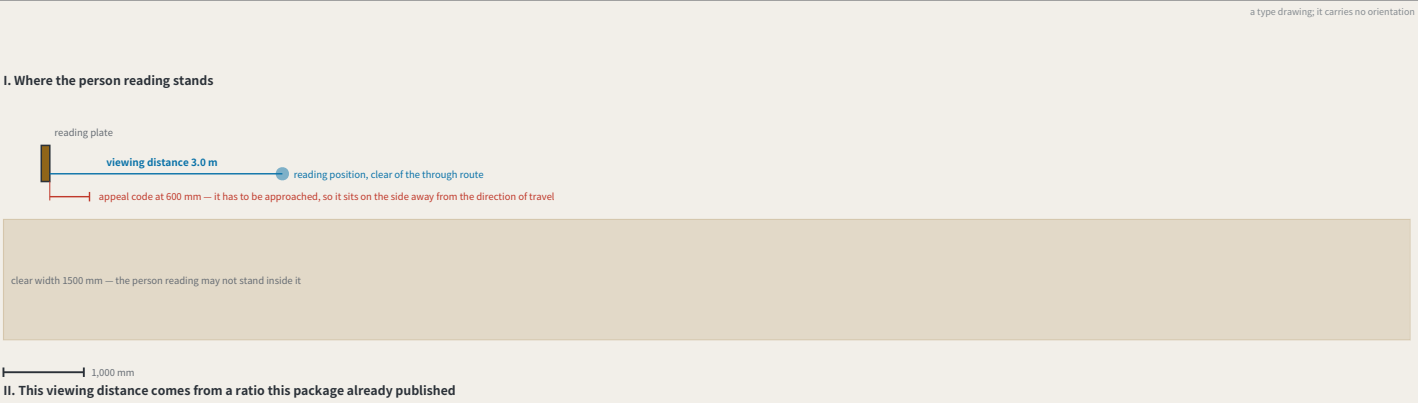
Item	Value	Why
Re-survey cycle	Monthly	third order is the most numerous tier and the one residents actually stand at
Legible cone	3.46 m	a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the distance FIG.27 fixes, 3.0 m, the cone opens to this width
Readers at once	5	cone ÷ a 600 mm shoulder. This is this sheet answering the question FIG.27 left open — that sheet fixed one person's position
A class of thirty	6 rounds, about 4 minutes	past that count the extra people are not handled by ground but by taking turns — and turns are a time cost, which belongs in writing rather than being discovered on site
Which way a queue runs	parallel to the kerb, never perpendicular to the direction of travel	a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not
What is not provided	no railing, no dedicated waiting area	a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure

The origin and the second order each have a sheet; the third order is the tier nobody drew, and the one residents actually stand at. It also inherits what FIG.27 left open: that sheet's 3.0 m distance is one person's position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — that part is solved by turns, not by ground.

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One reading: where a person stands, and whether they block the way

FIG.27



III. The four steps of one reading, and the ground each takes

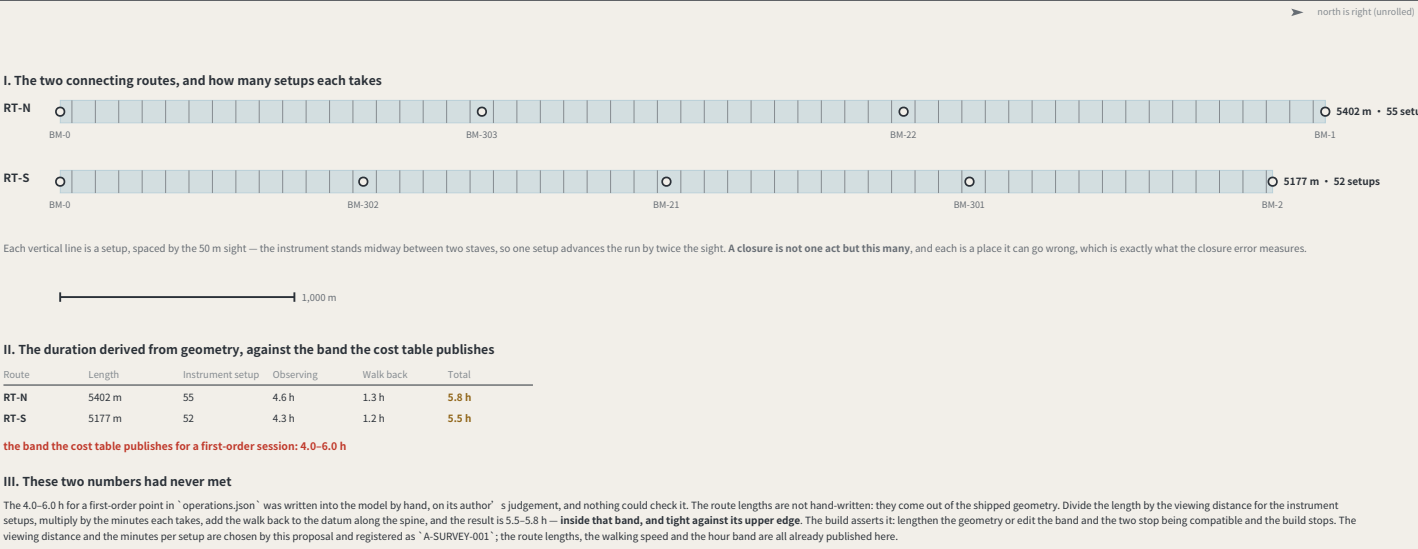
Step	What happens	Why that dimension
Walk there	FIG.21 measured the worst walk at 1,111 m — 30.9 minutes for PS — and this proposal does not meet its own rule	The cost of this step has been measured, and it measures as non-compliant
Stand	Reading at 3.0 m from the plate face, standing outside the clear walking width	Cap height 12 mm × viewing distance ratio 250 = 3.0 m; take the cap height as 6 mm and the viewing distance drops to 1.5 m, putting the person reading in the middle of the path
Read	Read the number on the plate and the benchmark_id	The number makes this reading attributable; no need to ask afterwards where the person was standing
Disagree	Scanning the appeal code requires approaching to about 600 mm	This step has to come within close range, so the stacking goes on the side facing away from the direction of travel — the one step that must intrude intrudes on the edge of the path, not its centre

The whole participation argument rests on a single sentence — anyone can walk up to a benchmark and take a reading — yet no drawing has ever drawn the four minutes that person spends standing there. Standing is not free: the ground he occupies is the ground others walk on. Working backwards from the "legible cap height ≈ viewing distance ÷ 250" already published in this package, a cap height of 12 mm gives a viewing distance of 3.0 m, which falls just outside the 1500 mm clear walking width; take 6 mm and the reader stands in the middle of the path. The cap height is set by this proposal and registered as A-PLATE-001.

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The annual first-order closure: how long a route actually takes

FIG.29



The first order was the last tier without a sheet, and it is not really a place but an event: once a year a connecting route is run, and this proposal's measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52–55 setups and 5.5–5.8 h — inside the 4.0–6.0 h the cost table sets by hand — and the build asserts the two stay compatible.

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The year: where forty-seven readings fall

FIG.30

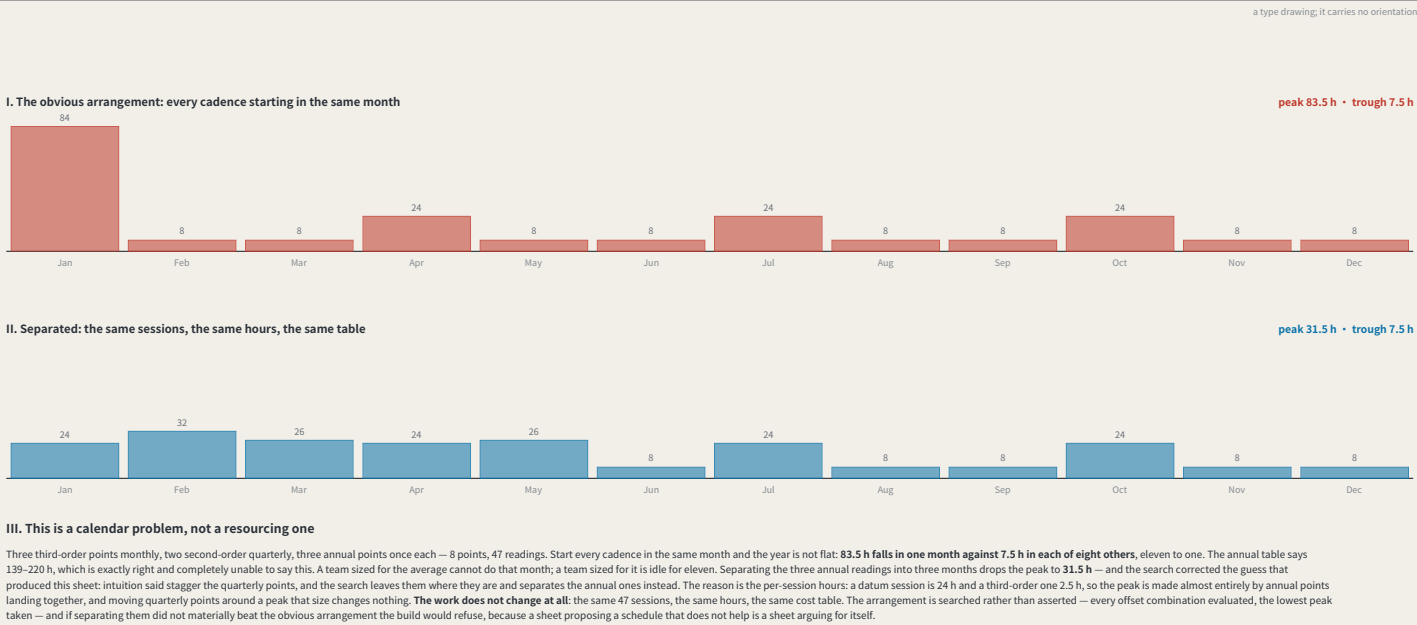


FIG.29 priced **one** closure and the cost table prices a **year** — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with **no change to the work at all**. This is a calendar problem, not a resourcing one.

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The other year: the hours this network asks of people it does not pay

FIG.31



FIG.30 flattened **paid** hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. Both cannot be minimised at once, and this proposal takes the volunteer peak: a paid peak is a procurement problem, a volunteer peak is a participation failure.

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How many people: the cheapest roster is the one that empties the instrument

FIG.32

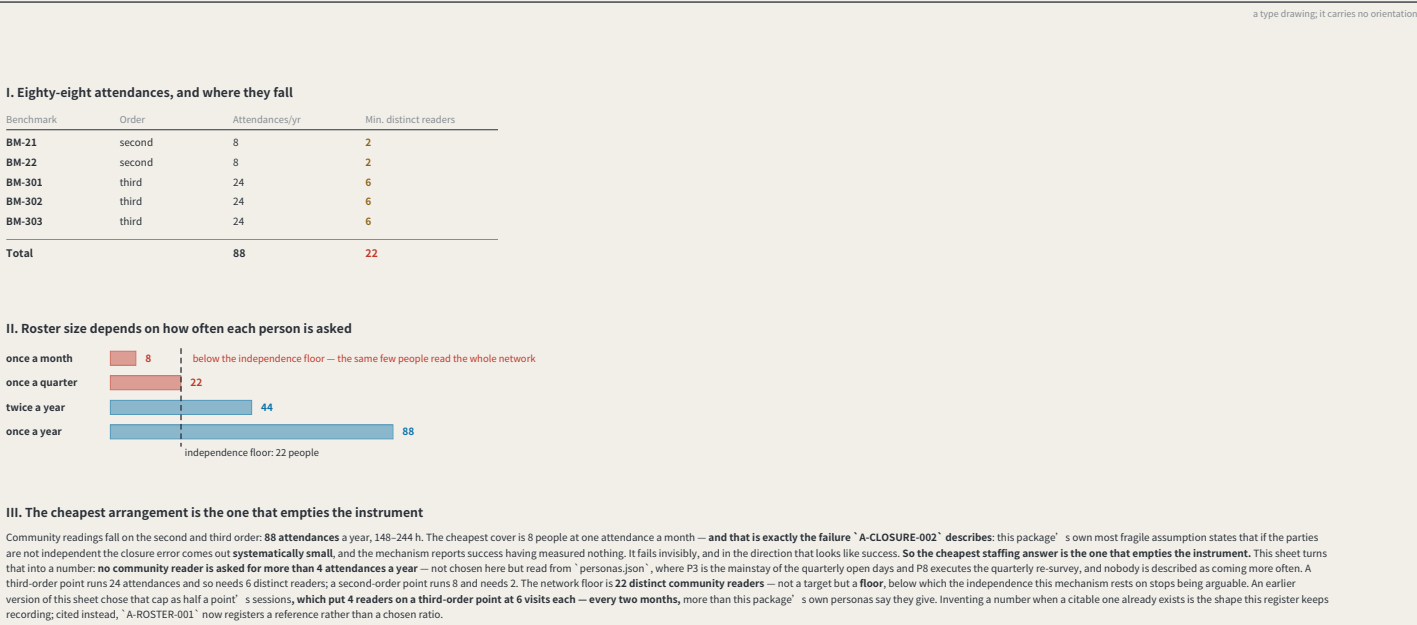
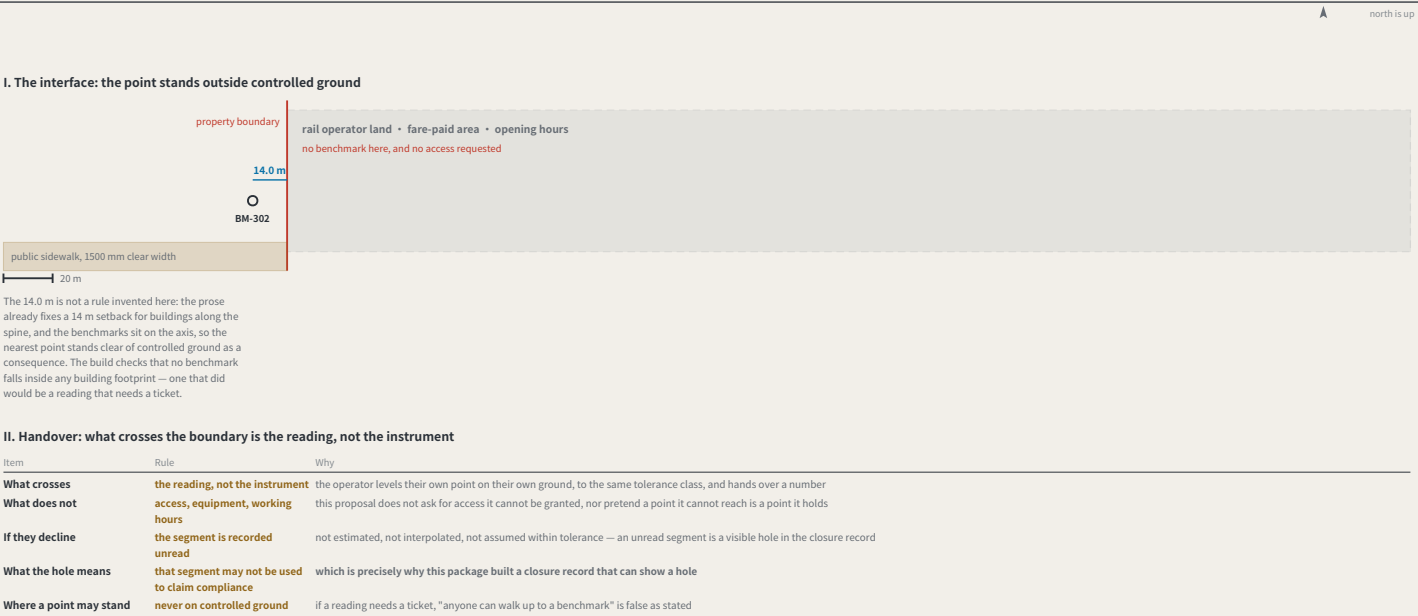


FIG.31 scheduled volunteer **hours** and never asked how many **people** that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — **which is exactly the failure A-CLOSURE-002 sets out**: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark’ s readings in a year, giving a network floor of 22 people.

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Dazhongsi: where the belt meets ground this network does not own

FIG.33



Every other sheet assumes this network reaches its own points; at the station that ends — the landowner is not the city. Three rules: **no benchmark on controlled ground** (BM-302 stands 14.0 m clear of the hub, checked at build time); **what crosses is the reading, not the instrument**; and **if the operator declines the segment is recorded unread** — not estimated, not interpolated, a visible hole in the closure record.

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